

```
import tarfile
import pandas as pd

file_path = "test-clean.tar.gz"

with tarfile.open(file_path, "r:gz") as tar:
    names = tar.getnames()
    i=0
    for j in names :
        print( j)
        i+=1
        if i==100:
            break
```

```
LibriSpeech/LICENSE.TXT
LibriSpeech/README.TXT
LibriSpeech/CHAPTERS.TXT
LibriSpeech/SPEAKERS.TXT
LibriSpeech/BOOKS.TXT
LibriSpeech/test-clean
LibriSpeech/test-clean/6930
LibriSpeech/test-clean/6930/75918
LibriSpeech/test-clean/6930/75918/6930-75918-0000.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0007.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0006.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0005.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0004.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0002.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0011.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0015.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0003.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0010.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0001.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0018.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0008.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0017.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0016.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0013.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0019.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0014.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0009.flac
LibriSpeech/test-clean/6930/75918/6930-75918-0020.flac
LibriSpeech/test-clean/6930/75918/6930-75918.trans.txt
LibriSpeech/test-clean/6930/75918/6930-75918-0012.flac
LibriSpeech/test-clean/6930/76324
LibriSpeech/test-clean/6930/76324/6930-76324-0006.flac
LibriSpeech/test-clean/6930/76324/6930-76324-0026.flac
LibriSpeech/test-clean/6930/76324/6930-76324-0015.flac
LibriSpeech/test-clean/6930/76324/6930-76324.trans.txt
```

[illegible]

```
LibriSpeech/test-clean/6930/81414/6930-81414-0026.flac
LibriSpeech/test-clean/6930/81414/6930-81414-0018.flac
LibriSpeech/test-clean/6930/81414/6930-81414-0013.flac
LibriSpeech/test-clean/6930/81414/6930-81414-0025.flac
LibriSpeech/test-clean/6930/81414/6930-81414-0017.flac
LibriSpeech/test-clean/6930/81414/6930-81414-0006.flac
LibriSpeech/test-clean/6930/81414/6930-81414-0015.flac
LibriSpeech/test-clean/1320
LibriSpeech/test-clean/1320/122617
LibriSpeech/test-clean/1320/122617/1320-122617-0003.flac
LibriSpeech/test-clean/1320/122617/1320-122617-0012.flac
LibriSpeech/test-clean/1320/122617/1320-122617-0026.flac
LibriSpeech/test-clean/1320/122617/1320-122617-0022.flac
LibriSpeech/test-clean/1320/122617/1320-122617-0002.flac
LibriSpeech/test-clean/1320/122617/1320-122617-0025.flac
LibriSpeech/test-clean/1320/122617/1320-122617-0036.flac
```

```
import os
```

```
print(os.path.getsize(file_path) / (1024 * 1024), "MB")
```

```
330.6045379638672 MB
```

```
!rm -rf *
```

```
with tarfile.open(file_path, "r:gz") as tar:
    tar.extractall("librispeech_test_clean")
```

```
print("Extraction complete")
```

```
/tmp/ipykernel_681/1454776725.py:2: DeprecationWarning: Python 3.14
will, by default, filter extracted tar archives and reject files or
modify their metadata. Use the filter argument to control this
behavior.
```

```
    tar.extractall("librispeech_test_clean")
```

```
Extraction complete
```

```
import os
```

```
for root, dirs, files in os.walk("librispeech_test_clean"):
    flac_files = [f for f in files if f.endswith(".flac")]
    trans_files = [f for f in files if f.endswith(".trans.txt")]
```

```
    if flac_files or trans_files:
        print("Folder:", root)
        print("Sample audio:", flac_files[:3])
        print("Transcript file:", trans_files[:3])
        break
```

Folder: librispeech\_test\_clean/LibriSpeech/test-clean/2830/3979  
Sample audio: ['2830-3979-0012.flac', '2830-3979-0005.flac', '2830-3979-0000.flac']  
Transcript file: ['2830-3979.trans.txt']

```
len(flac_files)

13

root_dir = "librispeech_test_clean/LibriSpeech/test-clean"

rows = []

for speaker_id in os.listdir(root_dir):
    speaker_path = os.path.join(root_dir, speaker_id)

    if not os.path.isdir(speaker_path):
        continue

    for chapter_id in os.listdir(speaker_path):
        chapter_path = os.path.join(speaker_path, chapter_id)

        if not os.path.isdir(chapter_path):
            continue

        trans_files = [f for f in os.listdir(chapter_path) if
f.endswith(".trans.txt")]

        for trans_file in trans_files:
            trans_path = os.path.join(chapter_path, trans_file)

            with open(trans_path, "r") as f:
                lines = f.readlines()

            for line in lines:
                parts = line.strip().split(" ", 1)
                audio_id = parts[0]
                transcript = parts[1]

                audio_path = os.path.join(chapter_path, audio_id +
".flac")

                if os.path.exists(audio_path):
                    rows.append({
                        "audio_path": audio_path,
                        "transcript": transcript
                    })

df = pd.DataFrame(rows)

df_400 = df.head(400)
```

```
df_400.to_csv("librispeech_300.csv", index=False)
```

```
print(df_400.head())
```

```
print("Total samples saved:", len(df_400))
```

```
          audio_path \
0  librispeech_test_clean/LibriSpeech/test-clean/...
1  librispeech_test_clean/LibriSpeech/test-clean/...
2  librispeech_test_clean/LibriSpeech/test-clean/...
3  librispeech_test_clean/LibriSpeech/test-clean/...
4  librispeech_test_clean/LibriSpeech/test-clean/...
```

```
          transcript
0  WE WANT YOU TO HELP US PUBLISH SOME LEADING WO...
1  THE CONDITION IS THAT I WILL BE PERMITTED TO M...
2  LET US BEGIN WITH THAT HIS COMMENTARY ON GALAT...
3  THE UNDERTAKING WHICH SEEMED SO ATTRACTIVE WHE...
4                      IT WAS WRITTEN IN LATIN
```

```
Total samples saved: 400
```

```
df.sample(5)
```

```
{
  "summary": {
    "\n  \"name\": \"df\",
    "\n  \"rows\": 5,
    "\n  \"fields\": [
      {
        "\n    \"column\": \"audio_path\",
        "\n    \"properties\": {
          "\n      \"dtype\": \"string\",
          "\n      \"num_unique_values\": 5,
          "\n      \"samples\": [
        "\n      \"librispeech_test_clean/LibriSpeech/test-clean/237/126133/237-126133-0021.flac\",
        "\n      \"librispeech_test_clean/LibriSpeech/test-clean/7176/92135/7176-92135-0006.flac\",
        "\n      \"librispeech_test_clean/LibriSpeech/test-clean/6829/68769/6829-68769-0007.flac\"
      ],
        "\n      \"semantic_type\": \"\",
        "\n      \"description\": \"\"
      }
    ],
    {
      "\n      \"column\": \"transcript\",
      "\n      \"properties\": {
        "\n          \"dtype\": \"string\",
          "\n          \"num_unique_values\": 5,
          "\n          \"samples\": [
        "\n          \"SHE ASKED IMPULSIVELY I DIDN'T BELIEVE YOU COULD PERSUADE HER FATHER\",
        "\n          \"I'VE GOT A LITTLE IDEA FOR A PLAY ABOUT A MAN AND A WOMAN AND ANOTHER WOMAN AND BUT PERHAPS I'D BETTER KEEP THE PLOT A SECRET FOR THE MOMENT\",
        "\n          \"BUT UNDER THE CIRCUMSTANCES I DOUBT IF SUCH AN ARRANGEMENT COULD BE MADE\"
        ],
        "\n          \"semantic_type\": \"\",
        "\n          \"description\": \"\"
        }
      }
    ]
  },
  \"type\": \"dataframe\"
}
```

```
df_400.shape
```

```
(400, 2)
```

```
from sklearn.model_selection import train_test_split
```

```
dev_df, eval_df = train_test_split(df_400, test_size= 0.2,
random_state= 42)
```

```
dev_df
```

```
{
  "summary": {
    "name": "dev_df",
    "rows": 320,
    "fields": [
      {
        "column": "audio_path",
        "properties": {
          "dtype": "string",
          "num_unique_values": 320,
          "samples": [
            "librispeech_test_clean/LibriSpeech/test-clean/8230/279154/8230-279154-0017.flac",
            "librispeech_test_clean/LibriSpeech/test-clean/2830/3980/2830-3980-0001.flac",
            "librispeech_test_clean/LibriSpeech/test-clean/2830/3979/2830-3979-0007.flac"
          ]
        }
      },
      {
        "column": "transcript",
        "properties": {
          "dtype": "string",
          "num_unique_values": 320,
          "samples": [
            "THERE MAY BE A SPECIFIC FEELING WHICH COULD BE CALLED THE FEELING OF PASTNESS ESPECIALLY WHERE IMMEDIATE MEMORY IS CONCERNED",
            "THEY SAID TO THE GALATIANS YOU HAVE NO RIGHT TO THINK HIGHLY OF PAUL",
            "MUCH LATER WHEN A FRIEND OF HIS WAS PREPARING AN EDITION OF ALL HIS LATIN WORKS HE REMARKED TO HIS HOME CIRCLE IF I HAD MY WAY ABOUT IT THEY WOULD REPUBLISH ONLY THOSE OF MY BOOKS WHICH HAVE DOCTRINE MY GALATIANS FOR INSTANCE"
          ]
        }
      }
    ]
  },
  "type": "dataframe",
  "variable_name": "dev_df"
}
```

```
dev_df.to_csv("dev_320.csv", index=False)
eval_df.to_csv("eval_80.csv", index=False)
```

```
print("Dev samples:", len(dev_df))
print("Eval samples:", len(eval_df))
print(eval_df.head())
```

```
Dev samples: 320
Eval samples: 80
```

```
audio_path \
209 librispeech_test_clean/LibriSpeech/test-clean/...
280 librispeech_test_clean/LibriSpeech/test-clean/...
33 librispeech_test_clean/LibriSpeech/test-clean/...
210 librispeech_test_clean/LibriSpeech/test-clean/...
93 librispeech_test_clean/LibriSpeech/test-clean/...
```

```
transcript
209 OF THIS SECOND LETTER ALSO SHE SPOKE AND TOLD ...
280 THE LURE PROVED TOO MUCH FOR HIM AND HE CAME S...
33 THIS IS NO SINFUL PRIDE IT IS HOLY PRIDE
210 ON AUGUST TWENTY SEVENTH EIGHTEEN THIRTY SEVEN...
93 AND WHAT SORT OF EVIDENCE IS LOGICALLY POSSIBLE
```

```
!pip install -q openai-whisper jiwer
```



```
sample_wer = wer(true_norm, pred_norm)
```

```
print("TRUE:", true_norm)
```

```
print("PRED:", pred_norm)
```

```
print("WER:", sample_wer)
```

```
TRUE: WE SHED OUR BLOOD FOR OUR COUNTRY
```

```
PRED: WE SHED OUR BLOOD FOR OUR COUNTRY
```

```
WER: 0.0
```

```
predictions = []
```

```
wers = []
```

```
for i, row in eval_df.iterrows():
```

```
    audio_path = row["audio_path"]
```

```
    true_text = row["transcript"]
```

```
    result = model.transcribe(audio_path)
```

```
    pred_text = result["text"]
```

```
    true_norm = true_text
```

```
    pred_norm = normalize_text(pred_text)
```

```
    sample_wer = wer(true_norm, pred_norm)
```

```
    predictions.append(pred_norm)
```

```
    wers.append(sample_wer)
```

```
eval_df["predicted_text"] = predictions
```

```
eval_df["wer"] = wers
```

```
print("Average WER:", sum(wers) / len(wers))
```

```
eval_df.head()
```

```
Average WER: 0.11394762704472346
```

```
{"summary": "{\n  \"name\": \"eval_df\",\n  \"rows\": 80,\n  \"fields\": [\n    {\n      \"column\": \"audio_path\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 80,\n        \"samples\": [\n          \"librispeech_test_other/LibriSpeech/test-other/4294/35475/4294-35475-0024.flac\",\n          \"librispeech_test_other/LibriSpeech/test-other/5442/41168/5442-41168-0004.flac\",\n          \"librispeech_test_other/LibriSpeech/test-other/4294/35475/4294-35475-0020.flac\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\",\n        \"column\": \"audio_path\",\n        \"transcript\": \"AMONG THOSE WHO DREW BACK WERE ETHELRIED'S BROTHERS THE THREE THAT WERE DARK AND THE THREE THAT WERE FAIR\",\n        \"properties\": {\n          \"dtype\": \"string\",\n          \"num_unique_values\": 80,\n          \"samples\": [\n            \"AMONG THOSE WHO DREW BACK WERE ETHELRIED'S BROTHERS THE THREE THAT WERE DARK AND THE THREE THAT WERE FAIR\",\n            \"WE SHED OUR BLOOD FOR OUR COUNTRY\",\n            \"WHAT IS TO BECOME OF ME\"
```



```

CRIED THE POOR PEASANT\\",\\n        ],\\n        \\\"semantic_type\\\":
\\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n        }\\n        },\\n        {\\n
\\\"column\\\": \\\"predicted_text\\\",\\n        \\\"properties\\\": {\\n
\\\"dtype\\\": \\\"string\\\",\\n        \\\"num_unique_values\\\": 80,\\n
\\\"samples\\\": [\\n        \\\"AMONG THOSE WHO DREW BACK WERE ETHELREDES
BROTHERS THE THREE THAT WERE DARK AND THE THREE THAT WERE FAIR\\\",\\n
\\\"WE SHED OUR BLOOD FOR OUR COUNTRY\\\",\\n        \\\"WHAT IS TO BECOME
OF ME CRIED THE POOR PEASANT\\\"\\n        ],\\n        \\\"semantic_type\\\":
\\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n        }\\n        },\\n        {\\n
\\\"column\\\": \\\"wer\\\",\\n        \\\"properties\\\": {\\n        \\\"dtype\\\":
\\\"number\\\",\\n        \\\"std\\\": 0.15064509085312722,\\n        \\\"min\\\":
0.0,\\n        \\\"max\\\": 1.0,\\n        \\\"num_unique_values\\\": 43,\\n
\\\"samples\\\": [\\n        0.044444444444444446,\\n
0.07142857142857142,\\n        0.05\\n        ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n        }\\
n        }\\n        ]\\n}\\\",\\\"type\\\":\\\"dataframe\\\",\\\"variable_name\\\":\\\"eval_df\\\"}

```

```

eval_df.to_csv("whisper_eval_results.csv", index=False)
print("Saved whisper_eval_results.csv")

```

Saved whisper\_eval\_results.csv

```

print("Lowest WER examples:")
display(eval_df.sort_values("wer").head(5))

```

```

print("Highest WER examples:")
display(eval_df.sort_values("wer", ascending=False).head(5))

```

Lowest WER examples:

```

{"summary": "{\\n  \\\"name\\\": \\\"display(eval_df\\\",\\n  \\\"rows\\\": 5,\\n
\\\"fields\\\": [\\n    {\\n      \\\"column\\\": \\\"audio_path\\\",\\n
\\\"properties\\\": {\\n        \\\"dtype\\\": \\\"string\\\",\\n
\\\"num_unique_values\\\": 5,\\n        \\\"samples\\\": [\\n
\\\"librispeech_test_other/LibriSpeech/test-other/4294/9934/4294-9934-
0013.flac\\\",\\n
\\\"librispeech_test_other/LibriSpeech/test-
other/3997/180297/3997-180297-0011.flac\\\",\\n
\\\"librispeech_test_other/LibriSpeech/test-other/8280/266249/8280-
266249-0009.flac\\\"\\n        ],\\n        \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n        }\\n        },\\n        {\\n        \\\"column\\\":
\\\"transcript\\\",\\n        \\\"properties\\\": {\\n        \\\"dtype\\\":
\\\"string\\\",\\n        \\\"num_unique_values\\\": 5,\\n        \\\"samples\\\":
[\\n        \\\"WHAT ARE YOU GOING TO DO I DO NOT KNOW\\\",\\n
\\\"WELL GOOD HEAVENS THE MEANS WERE EASY ENOUGH TO GUESS\\\",\\n
\\\"HE CERTAINLY LOOKS LIKE A VERY NICE LITTLE BOY\\\"\\n        ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n        }\\
n        },\\n        {\\n        \\\"column\\\": \\\"predicted_text\\\",\\n
\\\"properties\\\": {\\n        \\\"dtype\\\": \\\"string\\\",\\n
\\\"num_unique_values\\\": 5,\\n        \\\"samples\\\": [\\n        \\\"WHAT
ARE YOU GOING TO DO I DO NOT KNOW\\\",\\n        \\\"WELL GOOD HEAVENS

```

```
THE MEANS WERE EASY ENOUGH TO GUESS\",\\n          \\\"HE CERTAINLY LOOKS
LIKE A VERY NICE LITTLE BOY\"\\n          ],\\n          \\\"semantic_type\\\":
\\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\n          },\\n          {\\n
\\\"column\\\": \\\"wer\\\",\\n          \\\"properties\\\": {\\n          \\\"dtype\\\":
\\\"number\\\",\\n          \\\"std\\\": 0.0,\\n          \\\"min\\\": 0.0,\\n
\\\"max\\\": 0.0,\\n          \\\"num_unique_values\\\": 1,\\n          \\\"samples\\\":
[\\n          0.0\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n          }\\n          }\\n          ]\\n}\\",\"type\":\"dataframe\"}
```

Highest WER examples:

```
{\"summary\":{\"\\n  \\\"name\\\": \\\"display(eval_df\\\",\\n  \\\"rows\\\": 5,\\n
\\\"fields\\\": [\\n    {\\n      \\\"column\\\": \\\"audio_path\\\",\\n
\\\"properties\\\": {\\n      \\\"dtype\\\": \\\"string\\\",\\n
\\\"num_unique_values\\\": 5,\\n      \\\"samples\\\": [\\n
\\\"librispeech_test_other/LibriSpeech/test-other/4294/35475/4294-35475-
0025.flac\\\",\\n      \\\"librispeech_test_other/LibriSpeech/test-
other/3997/182399/3997-182399-0014.flac\\\",\\n
\\\"librispeech_test_other/LibriSpeech/test-other/3997/182399/3997-
182399-0016.flac\\\"\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n      },\\n      {\\n      \\\"column\\\":
\\\"transcript\\\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\":
\\\"string\\\",\\n      \\\"num_unique_values\\\": 5,\\n      \\\"samples\\\":
[\\n      \\\"BUT ETHELRIED HEEDED NOT THEIR TAUNTS\\\",\\n
\\\"BUT HE DON'T MIND THAT\\\",\\n      \\\"YES SAH SHE SHO'LY WAS PLUMB
SCARED\\\"\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n      },\\n      {\\n      \\\"column\\\":
\\\"predicted_text\\\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\":
\\\"string\\\",\\n      \\\"num_unique_values\\\": 5,\\n      \\\"samples\\\":
[\\n      \\\"BUT ETHEL READ HE DID NOT THEIR TAUNTS\\\",\\n
\\\"BUT YOU DONT MIND THAT\\\",\\n      \\\"YES AH SHE SURELY WAS
PLUMSCARED\\\"\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n      },\\n      {\\n      \\\"column\\\":
\\\"wer\\\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\": \\\"number\\\",\\n
\\\"std\\\": 0.1111616269485951,\\n      \\\"min\\\": 0.4,\\n      \\\"max\\\":
0.6666666666666666,\\n      \\\"num_unique_values\\\": 4,\\n
\\\"samples\\\": [\\n      0.5714285714285714,\\n      0.4,\\n
0.6666666666666666\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n      }\\n      ]\\n}\\",\"type\":\"dataframe\"}
```

this is a pretrained model and we would like to see the results when we train a wav2vec2 model somewhat from scratch

```
!pip install -q transformers datasets evaluate jiwer torchaudio
accelerate
```

```
===== 0.0/84.1 kB ? eta -:--:--
===== 84.1/84.1 kB 4.5 MB/s eta
```

```
0:00:00
0.0/3.1 MB ? eta -:-:-:-
2.3/3.1 MB 70.3 MB/s eta
0:00:01 3.1/3.1 MB 56.0 MB/s
eta 0:00:00
```

```
import numpy as np
```

```
import torch
```

```
print(torch.cuda.is_available())
```

```
True
```

```
from transformers import Wav2Vec2Processor, Wav2Vec2ForCTC
```

```
model_name = "facebook/wav2vec2-base-960h"
```

```
processor = Wav2Vec2Processor.from_pretrained(model_name)
```

```
w2v_model = Wav2Vec2ForCTC.from_pretrained(model_name)
```

```
w2v_model.to("cuda")
```

```
print("wav2vec2 loaded")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
```

```
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
```

```
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
```

```
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.
```

```
{"model_id": "c5c6c1dad4404037b4ee5e8849e29720", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "093ef35a9ead4f67b7b579417d190f49", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "fb79a25d3dd04af6ac5e2b10ebabae87", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "37bef5ce0538414eb9e4ce43f9eb83e8", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "d8e7063472ec47be99df24baff17f451", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "d043cfb9534b498b8576e414d6b1cb17", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "bb6e445fc8604ec5a1cdb370eec7d729", "version_major": 2, "version_minor": 0}
```

[transformers] Wav2Vec2ForCTC LOAD REPORT from: facebook/wav2vec2-base-960h

| Key                        | Status  |
|----------------------------|---------|
| wav2vec2.masked_spec_embed | MISSING |

Notes:

- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

wav2vec2 loaded

```
from datasets import Dataset
```

```
train_dataset = Dataset.from_pandas(dev_df.reset_index(drop=True))  
eval_dataset = Dataset.from_pandas(eval_df.reset_index(drop=True))
```

```
print(train_dataset)  
print(eval_dataset)
```

```
Dataset({  
  features: ['audio_path', 'transcript'],  
  num_rows: 320  
})  
Dataset({  
  features: ['audio_path', 'transcript'],  
  num_rows: 80  
})
```

```
import librosa
```

```
def load_audio(batch):  
    audio, sr = librosa.load(batch["audio_path"], sr=16000)  
    batch["speech"] = audio  
    batch["sampling_rate"] = sr  
    batch["target_text"] = batch["transcript"].upper()  
    return batch
```

```
train_dataset = train_dataset.map(load_audio)  
eval_dataset = eval_dataset.map(load_audio)
```

```
print(train_dataset[0].keys())
```

```

{"model_id": "c3ede50c9ee146db8c23a28809fe646d", "version_major": 2, "version_minor": 0}

{"model_id": "6d3e10325fdc47bca848644abf5beb36", "version_major": 2, "version_minor": 0}

dict_keys(['audio_path', 'transcript', 'speech', 'sampling_rate', 'target_text'])

def prepare_batch(batch):
    # audio input
    audio_inputs = processor(
        audio=batch["speech"],
        sampling_rate=16000
    )

    # text label
    label_inputs = processor(
        text=batch["target_text"]
    )

    batch["input_values"] = audio_inputs["input_values"][0]
    batch["labels"] = label_inputs["input_ids"]

    return batch

train_dataset = train_dataset.map(
    prepare_batch,
    remove_columns=train_dataset.column_names
)

eval_dataset = eval_dataset.map(
    prepare_batch,
    remove_columns=eval_dataset.column_names
)

print(train_dataset[0].keys())

{"model_id": "788a9a5cf0584b79ad555dce5b82a2ad", "version_major": 2, "version_minor": 0}

{"model_id": "bfac2c67130641c9841bdcbl3e8a3918", "version_major": 2, "version_minor": 0}

dict_keys(['input_values', 'labels'])

from dataclasses import dataclass
from typing import Dict, List, Union

@dataclass
class DataCollatorCTCWithPadding:

```

```

processor: object
padding: Union[bool, str] = True

def __call__(self, features):
    input_features = [
        {"input_values": feature["input_values"]}
        for feature in features
    ]

    label_features = [
        {"input_ids": feature["labels"]}
        for feature in features
    ]

    batch = self.processor.pad(
        input_features,
        padding=self.padding,
        return_tensors="pt"
    )

    labels_batch = self.processor.pad(
        labels=label_features,
        padding=self.padding,
        return_tensors="pt"
    )

    labels = labels_batch["input_ids"].masked_fill(
        labels_batch.attention_mask.ne(1), -100
    )

    batch["labels"] = labels

    return batch

data_collator = DataCollatorCTCWithPadding(processor=processor)

import evaluate

wer_metric = evaluate.load("wer")

def compute_metrics(pred):
    pred_logits = pred.predictions
    pred_ids = np.argmax(pred_logits, axis=-1)

    pred.label_ids[pred.label_ids == -100] =
processor.tokenizer.pad_token_id

    pred_str = processor.batch_decode(pred_ids)
    label_str = processor.batch_decode(pred.label_ids,
group_tokens=False)

```

```

wer_value = wer_metric.compute(
    predictions=pred_str,
    references=label_str
)

return {"wer": wer_value}

{"model_id": "a283f96ab93149428caab3e7729295c3", "version_major": 2, "version_minor": 0}

w2v_model.freeze_feature_encoder()

from transformers import TrainingArguments

training_args = TrainingArguments(
    output_dir="./wav2vec2-finetuned-librispeech-400",
    per_device_train_batch_size=4,
    per_device_eval_batch_size=4,
    gradient_accumulation_steps=2,
    eval_strategy="epoch",
    save_strategy="epoch",
    num_train_epochs=10,
    learning_rate=1e-5,
    warmup_steps=50,
    fp16=True,
    logging_steps=10,
    save_total_limit=2,
    report_to="none"
)

from transformers import Trainer

trainer = Trainer(
    model=w2v_model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=eval_dataset,
    processing_class=processor,
    data_collator=data_collator,
    compute_metrics=compute_metrics
)

print("Trainer ready")

Trainer ready

trainer.train()

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

```

```

{"model_id": "093cd7b6e5e64caeb2e7e028ccf3b8a1", "version_major": 2, "version_minor": 0}

{"model_id": "45fce096f7a84dddbfbb23ea69256505", "version_major": 2, "version_minor": 0}

{"model_id": "f44dd9ac4b454d959495b76b07227dd6", "version_major": 2, "version_minor": 0}

{"model_id": "e43c11bc54554311ace47e65e81b084b", "version_major": 2, "version_minor": 0}

{"model_id": "a278dfaf8ed243bbb99b5a3f77a88c38", "version_major": 2, "version_minor": 0}

{"model_id": "02b23247e9f340bb97d78b768db53878", "version_major": 2, "version_minor": 0}

```

```

-----
KeyboardInterrupt                                Traceback (most recent call
last)
/tmp/ipykernel_1747/4032920361.py in <cell line: 0>()
----> 1 trainer.train()

/usr/local/lib/python3.12/dist-packages/transformers/trainer.py in
train(self, resume_from_checkpoint, trial, ignore_keys_for_eval)
    1431         ctx = suppress_progressBars() if args.push_to_hub
else contextlib.nullcontext()
    1432         with ctx:
-> 1433             return inner_training_loop(
    1434                 args=args,
    1435                 resume_from_checkpoint=resume_from_checkpoint,

/usr/local/lib/python3.12/dist-packages/transformers/trainer.py in
_inner_training_loop(self, batch_size, args, resume_from_checkpoint,
trial, ignore_keys_for_eval)
    1513         for epoch in range(epochs_trained, num_train_epochs):
    1514             self.control =
self.callback_handler.on_epoch_begin(self.args, self.state,
self.control)
-> 1515             self._run_epoch(
    1516                 model=model,
    1517                 epoch=epoch,

/usr/local/lib/python3.12/dist-packages/transformers/trainer.py in
_run_epoch(self, model, epoch, train_dataloader, steps_in_epoch,
num_update_steps_per_epoch, trial, ignore_keys_for_eval, start_time,
resume_from_checkpoint, epochs_trained,
steps_trained_in_current_epoch)
    1814

```



```

1815         self.control =
self.callback_handler.on_epoch_end(self.args, self.state,
self.control)
-> 1816         self._maybe_log_save_evaluate(
1817             self._tr_loss,
1818             grad_norm,

/usr/local/lib/python3.12/dist-packages/transformers/trainer.py in
_maybe_log_save_evaluate(self, tr_loss, grad_norm, model, trial,
epoch, ignore_keys_for_eval, start_time, learning_rate)
2100
2101         if self.control.should_save:
-> 2102             self._save_checkpoint(model, trial)
2103             self.control =
self.callback_handler.on_save(self.args, self.state, self.control)
2104

/usr/local/lib/python3.12/dist-packages/transformers/trainer.py in
_save_checkpoint(self, model, trial)
3072         run_dir = self._get_output_dir(trial=trial)
3073         output_dir = os.path.join(run_dir, checkpoint_folder)
-> 3074         self.save_model(output_dir, _internal_call=True)
3075
3076         if (

/usr/local/lib/python3.12/dist-packages/transformers/trainer.py in
save_model(self, output_dir, _internal_call)
3825
3826         elif self.args.should_save:
-> 3827             self._save(output_dir)
3828
3829         # Push to the Hub when `save_model` is called by the
user.

/usr/local/lib/python3.12/dist-packages/transformers/trainer.py in
_save(self, output_dir, state_dict)
3855         )
3856         else:
-> 3857             self.model.save_pretrained(output_dir,
state_dict=state_dict)
3858
3859             if self.processing_class is not None:

/usr/local/lib/python3.12/dist-packages/transformers/modeling_utils.py
in save_pretrained(self, save_directory, is_main_process, state_dict,
push_to_hub, max_shard_size, variant, token, save_peft_format,
save_original_format, **kwargs)
3556         # so it's not possible for now....
3557         # Write the shard to disk
-> 3558         safe_save_file(shard_state_dict, filename,

```

```

metadata=metadata)
3559         # Cleanup the data before next loop (important
with offloading, so we don't blowup cpu RAM)
3560         del shard_state_dict

/usr/local/lib/python3.12/dist-packages/safetensors/torch.py in
save_file(tensors, filename, metadata)
321     """
322     keep_references_alive = [] # to avoid garbage collection
of temporary numpy arrays while we write to disk
--> 323     serialize_file(
324         _flatten_as_ptr(tensors, keep_references_alive),
filename, metadata=metadata
325     )

KeyboardInterrupt:

trainer.evaluate()

<IPython.core.display.HTML object>

{'eval_loss': 126.21223449707031, 'eval_wer': 0.03007518796992481}

eval_df.to_csv("whisper_eval_results.csv", index=False)
trainer.save_model("./wav2vec2_finetuned")
processor.save_pretrained("./wav2vec2_finetuned")

{"model_id": "c09b1f2bdadf4eedb72af8725bce1ff8", "version_major": 2, "version_minor": 0}

['./wav2vec2_finetuned/processor_config.json']

# 1) Evaluate once and store actual Wav2Vec2 WER
w2v_eval_metrics = trainer.evaluate()
wav2vec_avg_wer = w2v_eval_metrics["eval_wer"]
print("Fine-tuned Wav2Vec2 Average WER:", wav2vec_avg_wer)

# 2) Save model and processor
trainer.save_model("./wav2vec2_finetuned")
processor.save_pretrained("./wav2vec2_finetuned")
print("Saved model to ./wav2vec2_finetuned")

# 3) Make per-sample Wav2Vec2 predictions and WER column

<IPython.core.display.HTML object>

Fine-tuned Wav2Vec2 Average WER: 0.03007518796992481

{"model_id": "48955cd48947479d8e548d3d4cffaf29", "version_major": 2, "version_minor": 0}

```

```

Saved model to ./wav2vec2_finetuned

w2v_model.eval()

w2v_predictions = []
w2v_wers = []

def transcribe_wav2vec(audio_path):
    speech, sr = librosa.load(audio_path, sr=16000)
    inputs = processor(speech, sampling_rate=16000,
return_tensors="pt", padding=True)
    input_values = inputs.input_values.to(w2v_model.device)

    with torch.no_grad():
        logits = w2v_model(input_values).logits

    pred_ids = torch.argmax(logits, dim=-1)
    pred_text = processor.batch_decode(pred_ids)[0]
    return pred_text

for _, row in eval_df.iterrows():
    true_text = row["transcript"].upper()
    pred_text = transcribe_wav2vec(row["audio_path"]).upper()

    sample_wer = wer(true_text, pred_text)
    w2v_predictions.append(pred_text)
    w2v_wers.append(sample_wer)

eval_df["wav2vec_predicted_text"] = w2v_predictions
eval_df["wav2vec_wer"] = w2v_wers

eval_df.to_csv("wav2vec_eval_results.csv", index=False)
print("Saved wav2vec_eval_results.csv")
print("Average per-sample Wav2Vec2 WER:", sum(w2v_wers) /
len(w2v_wers))
display(eval_df[["audio_path", "transcript", "wav2vec_predicted_text",
"wav2vec_wer"]].head())

wav2vec_avg_wer = sum(w2v_wers) / len(w2v_wers)

results = {
    "Model": ["Fine-tuned Wav2Vec2"],
    "Dataset": ["LibriSpeech eval split"],
    "WER": [wav2vec_avg_wer]
}

comparison_df = pd.DataFrame(results)
display(comparison_df)

Saved wav2vec_eval_results.csv
Average per-sample Wav2Vec2 WER: 0.025520290585938427

```

```

{"summary":{"\n  \"name\": \"display(comparison_df)\",\n  \"rows\": 5,\n  \"fields\": [\n    {\n      \"column\": \"audio_path\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 5,\n        \"samples\": [\n          \"librispeech_test_clean/LibriSpeech/test-clean/6930/76324/6930-76324-0016.flac\",\n          \"librispeech_test_clean/LibriSpeech/test-clean/8230/279154/8230-279154-0003.flac\",\n          \"librispeech_test_clean/LibriSpeech/test-clean/2830/3980/2830-3980-0020.flac\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\",\n        \"transcript\": \"THE LURE PROVED TOO MUCH FOR HIM AND HE CAME SPORTING AFTER IT AS FRISKILY AS A YOUNG KITTEN MUCH TO CYNTHIA'S DELIGHT WHEN SHE CAUGHT SIGHT OF HIM\",\n        \"properties\": {\n          \"dtype\": \"string\",\n          \"num_unique_values\": 5,\n          \"samples\": [\n            \"THE LURE PROVED TOO MUCH FOR HIM AND HE CAME SPORTING AFTER IT AS FRISKILY AS A YOUNG KITTEN MUCH TO CYNTHIA'S DELIGHT WHEN SHE CAUGHT SIGHT OF HIM\",\n            \"AND WHAT SORT OF EVIDENCE IS LOGICALLY POSSIBLE\",\n            \"THIS IS NO SINFUL PRIDE IT IS HOLY PRIDE\"\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\",\n          \"wav2vec_predicted_text\": \"\",\n          \"properties\": {\n            \"dtype\": \"string\",\n            \"num_unique_values\": 5,\n            \"samples\": [\n              \"THE LURE PROVED TOO MUCH FOR HIM AND HE CAME SPORTING AFTER IT AS FRISKILY AS A YOUNG KITTEN MUCH TO CYNTHIA'S DELIGHT WHEN SHE CAUGHT SIGHT OF HIM\",\n              \"AND WHAT SORT OF EVIDENCE IS LOGICALLY POSSIBLE\",\n              \"THIS IS NO SINFUL PRIDE IT IS HOLY PRIDE\"\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\",\n            \"wav2vec_wer\": \"\",\n            \"properties\": {\n              \"number\": \"std\": 0.0,\n              \"min\": 0.0,\n              \"max\": 0.0,\n              \"num_unique_values\": 1,\n              \"samples\": [\n                0.0\n              ],\n              \"semantic_type\": \"\",\n              \"description\": \"\"\n            }\n          }\n        ],\n        \"type\": \"dataframe\"}\n  ]\n}

{"summary":{"\n  \"name\": \"comparison_df\",\n  \"rows\": 1,\n  \"fields\": [\n    {\n      \"column\": \"Model\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 1,\n        \"samples\": [\n          \"Fine-tuned Wav2Vec2\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\",\n        \"Dataset\": \"\",\n        \"properties\": {\n          \"dtype\": \"string\",\n          \"num_unique_values\": 1,\n          \"samples\": [\n            \"LibriSpeech eval split\"\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\",\n          \"column\": \"WER\",\n          \"properties\": {\n            \"dtype\": \"number\",\n            \"std\": null,\n            \"min\": 0.025520290585938427,\n            \"max\": 0.025520290585938427,\n            \"num_unique_values\": 1,\n            \"samples\": [\n              0.025520290585938427\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          }\n        ]\n      }\n    ]\n  },\n  \"type\": \"dataframe\", \"variable_name\": \"comparison_df\"}

```

```

%%writefile app.py
import streamlit as st
import torch
import librosa
from transformers import Wav2Vec2ForCTC, Wav2Vec2Processor

MODEL_PATH = "wav2vec2_finetuned"

@st.cache_resource
def load_model():
    processor = Wav2Vec2Processor.from_pretrained(MODEL_PATH)
    model = Wav2Vec2ForCTC.from_pretrained(MODEL_PATH)
    model.eval()
    return processor, model

processor, model = load_model()

st.title("Speech to Text using Fine-tuned Wav2Vec2")

audio_file = st.file_uploader("Upload audio", type=["wav", "mp3", "flac"])

if audio_file is not None:
    st.audio(audio_file)

    speech, sr = librosa.load(audio_file, sr=16000)

    inputs = processor(
        speech,
        sampling_rate=16000,
        return_tensors="pt",
        padding=True
    )

    with torch.no_grad():
        logits = model(inputs.input_values).logits

    pred_ids = torch.argmax(logits, dim=-1)
    transcription = processor.batch_decode(pred_ids)[0]

    st.subheader("Transcription")
    st.write(transcription)

```

Writing app.py

```

!pip install streamlit librosa transformers torch -q
!kill -f streamlit

```

```
!streamlit run app.py --server.port 8501 --server.headless true --
server.enableCORS false --server.enableXsrfProtection false & npx
localtunnel --port 8501
```

on-worms-roll.local.lt

2026-06-20 14:29:25.929 Uvicorn server started on 0.0.0.0:8501

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>

Network URL: <http://172.28.0.12:8501>

External URL: <http://34.186.30.255:8501>

Loading weights: 100% 213/213 [00:00<00:00, 5603.52it/s]

[transformers] Accessing `__path__` from  
`models.aria.image_processing_aria`. Returning `__path__` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `__path__` from  
`models.aria.image_processing_pil_aria`. Returning `__path__`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `__path__` from  
`models.auto.image_processing_auto`. Returning `__path__` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `__path__` from  
`models.beit.image_processing_beit`. Returning `__path__` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `__path__` from  
`models.beit.image_processing_pil_beit`. Returning `__path__`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `__path__` from  
`models.bit.image_processing_bit`. Returning `__path__` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `__path__` from  
`models.bit.image_processing_pil_bit`. Returning `__path__` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `__path__` from  
`models.blip.image_processing_blip`. Returning `__path__` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `__path__` from  
`models.blip.image_processing_pil_blip`. Returning `__path__`  
instead. Behavior may be different and this alias will be removed in

future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.bridgetower.image\_processing\_bridgetower`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.bridgetower.image\_processing\_pil\_bridgetower`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.chameleon.image\_processing\_chameleon`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.chameleon.image\_processing\_pil\_chameleon`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.chinese\_clip.image\_processing\_chinese\_clip`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.chinese\_clip.image\_processing\_pil\_chinese\_clip`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.chmv2.image\_processing\_chmv2`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.clip.image\_processing\_clip`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.clip.image\_processing\_pil\_clip`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.cohere2\_vision.image\_processing\_cohere2\_vision`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.conditional\_detr.image\_processing\_conditional\_detr`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias  
will be removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.conditional\_detr.image\_processing\_pil\_conditional\_detr`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias  
will be removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from

``models.convnext.image_processing_convnext``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.convnext.image_processing_pil_convnext``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deepseek_ocr2.image_processing_deepseek_ocr2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deepseek_ocr2.image_processing_pil_deepseek_ocr2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deepseek_vl.image_processing_deepseek_vl``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deepseek_vl.image_processing_pil_deepseek_vl``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deepseek_vl_hybrid.image_processing_deepseek_vl_hybrid``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deepseek_vl_hybrid.image_processing_pil_deepseek_vl_hybrid``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deformable_detr.image_processing_deformable_detr``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deformable_detr.image_processing_pil_deformable_detr``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deit.image_processing_deit``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.deit.image_processing_pil_deit``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.depth_pro.image_processing_depth_pro``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in



future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.detr.image\_processing\_detr`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.detr.image\_processing\_pil\_detr`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.dinov3\_vit.image\_processing\_dinov3\_vit`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.donut.image\_processing\_donut`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.donut.image\_processing\_pil\_donut`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.dpt.image\_processing\_dpt`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.dpt.image\_processing\_pil\_dpt`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.efficientlofr.image\_processing\_efficientlofr`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.efficientlofr.image\_processing\_pil\_efficientlofr`.  
Returning `\_\_path\_\_` instead. Behavior may be different and this alias  
will be removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.efficientnet.image\_processing\_efficientnet`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.efficientnet.image\_processing\_pil\_efficientnet`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.emu3.image\_processing\_emu3`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.  
[transformers] Accessing `\_\_path\_\_` from

``models.eomt.image_processing_eomt``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.eomt.image_processing_pil_eomt``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.ernie4_5_vl_moe.image_processing_ernie4_5_vl_moe``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.ernie4_5_vl_moe.image_processing_pil_ernie4_5_vl_moe``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.flava.image_processing_flava``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.flava.image_processing_pil_flava``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.fuyu.image_processing_fuyu``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.fuyu.image_processing_pil_fuyu``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.gemma3.image_processing_gemma3``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.gemma3.image_processing_pil_gemma3``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.gemma4.image_processing_gemma4``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.gemma4.image_processing_pil_gemma4``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.gemma4_unified.image_processing_gemma4_unified``. Returning ``__path__`` instead. Behavior may be different and this alias will be

removed in future versions.

[transformers] Accessing `__path__` from ``.models.glm46v.image_processing_glm46v``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.glm46v.image_processing_pil_glm46v``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.glm4v.image_processing_glm4v``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.glm4v.image_processing_pil_glm4v``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.glm_image.image_processing_glm_image``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.glm_image.image_processing_pil_glm_image``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.glmga.image_processing_glmga``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.glmga.image_processing_pil_glmga``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.glpn.image_processing_glpn``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.glpn.image_processing_pil_glpn``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.got_ocr2.image_processing_got_ocr2``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.got_ocr2.image_processing_pil_got_ocr2``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from

``models.grounding_dino.image_processing_grounding_dino``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.grounding_dino.image_processing_pil_grounding_dino``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.idefics.image_processing_idefics``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.idefics.image_processing_pil_idefics``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.idefics2.image_processing_idefics2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.idefics2.image_processing_pil_idefics2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.idefics3.image_processing_idefics3``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.idefics3.image_processing_pil_idefics3``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.imagegpt.image_processing_imagegpt``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.imagegpt.image_processing_pil_imagegpt``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.janus.image_processing_janus``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.janus.image_processing_pil_janus``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.kosmos2_5.image_processing_kosmos2_5``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in

future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.kosmos2\_5.image\_processing\_pil\_kosmos2\_5`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.layoutlmv2.image\_processing\_layoutlmv2`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.layoutlmv2.image\_processing\_pil\_layoutlmv2`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.layoutlmv3.image\_processing\_layoutlmv3`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.layoutlmv3.image\_processing\_pil\_layoutlmv3`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.levit.image\_processing\_levit`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.levit.image\_processing\_pil\_levit`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.lfm2\_vl.image\_processing\_lfm2\_vl`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.lightglue.image\_processing\_lightglue`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.lightglue.image\_processing\_pil\_lightglue`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.llama4.image\_processing\_llama4`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.llava.image\_processing\_llava`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.  
[transformers] Accessing `\_\_path\_\_` from

``models.llava.image_processing_pil_llava``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.llava_next.image_processing_llava_next``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.llava_next.image_processing_pil_llava_next``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.llava_onevision.image_processing_llava_onevision``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.llava_onevision.image_processing_pil_llava_onevision``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.mask2former.image_processing_mask2former``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.mask2former.image_processing_pil_mask2former``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.maskformer.image_processing_maskformer``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.maskformer.image_processing_pil_maskformer``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.minicpmv4_6.image_processing_minicpmv4_6``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.minicpmv4_6.image_processing_pil_minicpmv4_6``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.mllama.image_processing_mllama``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.mllama.image_processing_pil_mllama``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in

future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.mobilenet\_v1.image\_processing\_mobilenet\_v1`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.mobilenet\_v1.image\_processing\_pil\_mobilenet\_v1`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.mobilenet\_v2.image\_processing\_mobilenet\_v2`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.mobilenet\_v2.image\_processing\_pil\_mobilenet\_v2`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.mobilevit.image\_processing\_mobilevit`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.mobilevit.image\_processing\_pil\_mobilevit`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.nougat.image\_processing\_nougat`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.nougat.image\_processing\_pil\_nougat`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.oneformer.image\_processing\_oneformer`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.oneformer.image\_processing\_pil\_oneformer`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.ovis2.image\_processing\_ovis2`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.ovis2.image\_processing\_pil\_ovis2`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from

``models.owlv2.image_processing_owlv2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.owlv2.image_processing_pil_owlv2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.owlvit.image_processing_owlvit``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.owlvit.image_processing_pil_owlvit``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.paddleocr_vl.image_processing_paddleocr_vl``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.paddleocr_vl.image_processing_pil_paddleocr_vl``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.perceiver.image_processing_perceiver``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.perceiver.image_processing_pil_perceiver``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.perception_lm.image_processing_perception_lm``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.phi4_multimodal.image_processing_phi4_multimodal``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.pi0.image_processing_pi0``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.pix2struct.image_processing_pil_pix2struct``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.pix2struct.image_processing_pix2struct``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in



future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pixtral.image\_processing\_pil\_pixtral`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pixtral.image\_processing\_pixtral`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.poolformer.image\_processing\_pil\_poolformer`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.poolformer.image\_processing\_poolformer`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pp\_chart2table.image\_processing\_pil\_pp\_chart2table`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias  
will be removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pp\_chart2table.image\_processing\_pp\_chart2table`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pp\_doclayout\_v2.image\_processing\_pp\_doclayout\_v2`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pp\_doclayout\_v3.image\_processing\_pp\_doclayout\_v3`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pp\_formulanet.image\_processing\_pp\_formulanet`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias will be  
removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pp\_lcnet.image\_processing\_pp\_lcnet`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pp\_ocrv5\_server\_det.image\_processing\_pp\_ocrv5\_server\_det`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias  
will be removed in future versions.  
[transformers] Accessing `\_\_path\_\_` from  
`.models.pp\_ocrv5\_server\_rec.image\_processing\_pp\_ocrv5\_server\_rec`. Returning  
`\_\_path\_\_` instead. Behavior may be different and this alias  
will be removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.pp\_ocrv6\_small\_rec.image\_processing\_pp\_ocrv6\_small\_rec`.  
Returning `\_\_path\_\_` instead. Behavior may be different and this alias  
will be removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.prompt\_depth\_anything.image\_processing\_pil\_prompt\_depth\_anyth  
ing`. Returning `\_\_path\_\_` instead. Behavior may be different and this  
alias will be removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.prompt\_depth\_anything.image\_processing\_prompt\_depth\_anything`  
. Returning `\_\_path\_\_` instead. Behavior may be different and this  
alias will be removed in future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.pvt.image\_processing\_pil\_pvt`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.pvt.image\_processing\_pvt`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.qwen2\_vl.image\_processing\_pil\_qwen2\_vl`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.qwen2\_vl.image\_processing\_qwen2\_vl`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.rf\_detr.image\_processing\_rf\_detr`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.rt\_detr.image\_processing\_pil\_rt\_detr`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.rt\_detr.image\_processing\_rt\_detr`. Returning `\_\_path\_\_`  
instead. Behavior may be different and this alias will be removed in  
future versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.sam.image\_processing\_pil\_sam`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `\_\_path\_\_` from  
`.models.sam.image\_processing\_sam`. Returning `\_\_path\_\_` instead.  
Behavior may be different and this alias will be removed in future  
versions.

[transformers] Accessing `\_\_path\_\_` from

``models.sam2.image_processing_sam2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.sam3.image_processing_sam3``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.sapiens2.image_processing_sapiens2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.segformer.image_processing_pil_segformer``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.segformer.image_processing_segformer``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.seggpt.image_processing_pil_seggpt``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.seggpt.image_processing_seggpt``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.siglip.image_processing_pil_siglip``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.siglip.image_processing_siglip``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.siglip2.image_processing_pil_siglip2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.siglip2.image_processing_siglip2``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.slanext.image_processing_slanext``. Returning ``__path__`` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing ``__path__`` from ``models.smolvlm.image_processing_pil_smolvlm``. Returning ``__path__``

instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.smolvlm.image_processing_smolvlm`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.superglue.image_processing_pil_superglue`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.superglue.image_processing_superglue`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.superpoint.image_processing_pil_superpoint`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.superpoint.image_processing_superpoint`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.swin2sr.image_processing_pil_swin2sr`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.swin2sr.image_processing_swin2sr`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.textnet.image_processing_pil_textnet`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.textnet.image_processing_textnet`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.timm_wrapper.image_processing_timm_wrapper`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.tvp.image_processing_pil_tvp`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from `models.tvp.image_processing_tvp`. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

versions.

[transformers] Accessing `__path__` from ``.models.uvdoc.image_processing_uvdoc``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.video_llama_3.image_processing_pil_video_llama_3``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.video_llama_3.image_processing_video_llama_3``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.video_llava.image_processing_video_llava``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.videomae.image_processing_pil_videomae``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.videomae.image_processing_videomae``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.vilt.image_processing_pil_vilt``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

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[transformers] Accessing `__path__` from ``.models.vit.image_processing_pil_vit``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.vit.image_processing_vit``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.vitmatte.image_processing_pil_vitmatte``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

[transformers] Accessing `__path__` from ``.models.vitmatte.image_processing_vitmatte``. Returning `__path__` instead. Behavior may be different and this alias will be removed in future versions.

```
[transformers] Accessing `__path__` from
`.models.vitpose.image_processing_pil_vitpose`. Returning `__path__`
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[transformers] Accessing `__path__` from
`.models.vitpose.image_processing_vitpose`. Returning `__path__`
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future versions.
[transformers] Accessing `__path__` from
`.models.vivit.image_processing_vivit`. Returning `__path__` instead.
Behavior may be different and this alias will be removed in future
versions.
[transformers] Accessing `__path__` from
`.models.yolos.image_processing_pil_yolos`. Returning `__path__`
instead. Behavior may be different and this alias will be removed in
future versions.
[transformers] Accessing `__path__` from
`.models.yolos.image_processing_yolos`. Returning `__path__` instead.
Behavior may be different and this alias will be removed in future
versions.
[transformers] Accessing `__path__` from
`.models.zoedepth.image_processing_pil_zoedepth`. Returning `__path__`
instead. Behavior may be different and this alias will be removed in
future versions.
[transformers] Accessing `__path__` from
`.models.zoedepth.image_processing_zoedepth`. Returning `__path__`
instead. Behavior may be different and this alias will be removed in
future versions.
Stopping...
```

^C

```
!pip install gradio -q
```

```
import gradio as gr
import torch
import librosa
from transformers import Wav2Vec2ForCTC, Wav2Vec2Processor
```

```
MODEL_PATH = "wav2vec2_finetuned"
```

```
processor = Wav2Vec2Processor.from_pretrained(MODEL_PATH)
model = Wav2Vec2ForCTC.from_pretrained(MODEL_PATH)
model.eval()
```

```
def transcribe(audio_path):
    speech, sr = librosa.load(audio_path, sr=16000)
    inputs = processor(speech, sampling_rate=16000,
    return_tensors="pt", padding=True)

    with torch.no_grad():
```

```
logits = model(inputs.input_values).logits
pred_ids = torch.argmax(logits, dim=-1)
return processor.batch_decode(pred_ids)[0]

demo = gr.Interface(
    fn=transcribe,
    inputs=gr.Audio(type="filepath"),
    outputs="text",
    title="Speech to Text using Fine-tuned Wav2Vec2"
)

demo.launch(share=True)

{"model_id": "70ccc490dc1a40f7b752a5ffa0f60b16", "version_major": 2, "version_minor": 0}
```

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

\* Running on public URL: <https://58e3ecdald888dd635.gradio.live>

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory to deploy to Hugging Face Spaces (<https://huggingface.co/spaces>)

<IPython.core.display.HTML object>